Lesson notes · for you

Test it honestly

AI · Level 3 · Lesson 3

Time
23 min

Materials
Paper and a pen; Ten made-up ★ sums that follow one secret rule: five to memorise and five new ones; A friend or family member to test

Learning goals

- Keep test data apart from training data, and count right and wrong answers.
- Explain overfitting: perfect on training examples, poor on new ones.
- Tell a false positive from a false negative, and say which matters more for your model.

Activity: Memoriser and learner

1. Set up (3 min): Make up a secret rule for a new sign, such as 3 ★ 4 = 13 (multiply, then add 1), and write ten ★ sums with their answers. Keep five as “training” sums and hide the other five.
2. Train (5 min): Player 1 memorises the five answers and must not look for the rule. Player 2 is told the rule.
3. Test (5 min): Ask both players the five training sums, then the five hidden sums they have never seen.
4. Score (4 min): Write each player’s training score and test score side by side, like the two bars in the video.
5. Explain (6 min): Which player overfitted? Write one sentence linking the memoriser to a model that is perfect on its training data, and one on which mistake would matter more in a malaria test.

Works with three people (two players and a quizmaster) or at home with family. Keep the hidden sums out of sight until the test.

Discussion

- Why is a score on training examples not an honest test?
- Your model is perfect on training photos and poor on new ones. What would you try?
- For a model that spots smoke in photos to warn of fire, which mistake is worse?

Check yourself

- My own example of a false alarm and a miss, and which one costs more.

Try it this week
Make up a secret rule for a new sign, such as 3 ★ 4 = 13 (multiply, then add 1), and write five ★ sums with their answers. Ask a friend to memorise the five answers without hunting for the rule, test them on the same five, then give them five new ★ sums. If they only memorised, they’ll struggle: explain overfitting to them using what just happened.

Worksheet

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Name _____ Class _____ Date _____

1 **Your model spots malaria in microscope photos of blood. Which mistake matters more?**

◯ Missing a real case ◯ They matter exactly the same ◯ A false alarm on someone healthy

2 **Your test set has 20 photos, but 8 of them are second photos of leaves that are also in the training set. Its score is still an honest measure.**

◯ True ◯ False

3 **A model looks for sick leaves. Match each result to its name.**

Says “sick”; the leaf is healthy	• • • •	Right: a sick leaf caught
Says “healthy”; the leaf is sick	• • • •	Right: a healthy leaf passed
Says “sick”; the leaf is sick	• • • •	False positive: a false alarm
Says “healthy”; the leaf is healthy	• • • •	False negative: a miss

4 **Complete the sentence.**

A model that is perfect on its _____ examples but poor on _____ ones is _____.

Word bank: overfitting · training · underfitting · balanced · new

Think of a model you might build. Which would cost more for it, a false alarm or a miss, and who would pay for it?

Check your answers

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1 **Missing a real case**

A missed case leaves a sick person untreated, and malaria can kill.

2 **False**

A second photo of the same leaf isn't really new: the model may recognise that leaf rather than the disease, so the score looks better than it will in a farmer's field.

3 **Says "sick"; the leaf is healthy — False positive: a false alarm**
Says "healthy"; the leaf is sick — False negative: a miss
Says "sick"; the leaf is sick — Right: a sick leaf caught
Says "healthy"; the leaf is healthy — Right: a healthy leaf passed

"Positive" means the model says yes, here "sick", and it is false when the leaf is really healthy.

4 **A model that is perfect on its training examples but poor on new ones is overfitting.**

That is overfitting: it memorised its training examples.